



# WorkshopPLUS: SharePoint Server Administration

SharePoint Migration Tool



# CONDITIONS AND TERMS OF USE:

© Microsoft Corporation. All rights reserved.

You may use these training materials solely for your personal internal reference and non-commercial purposes. You may not distribute, transmit, resell or otherwise make these training materials available to any other person or party without express permission from Microsoft Corporation. URL's or other internet website references in the training materials may change without notice. Unless otherwise noted, any companies, organizations, domain names, e-mail addresses, people, places and events depicted in the training materials are for illustration only and are fictitious. No real association is intended or inferred. THESE TRAINING MATERIALS ARE PROVIDED "AS IS"; MICROSOFT MAKES NO WARRANTIES, EXPRESS OR IMPLIED IN THESE TRAINING MATERIALS.

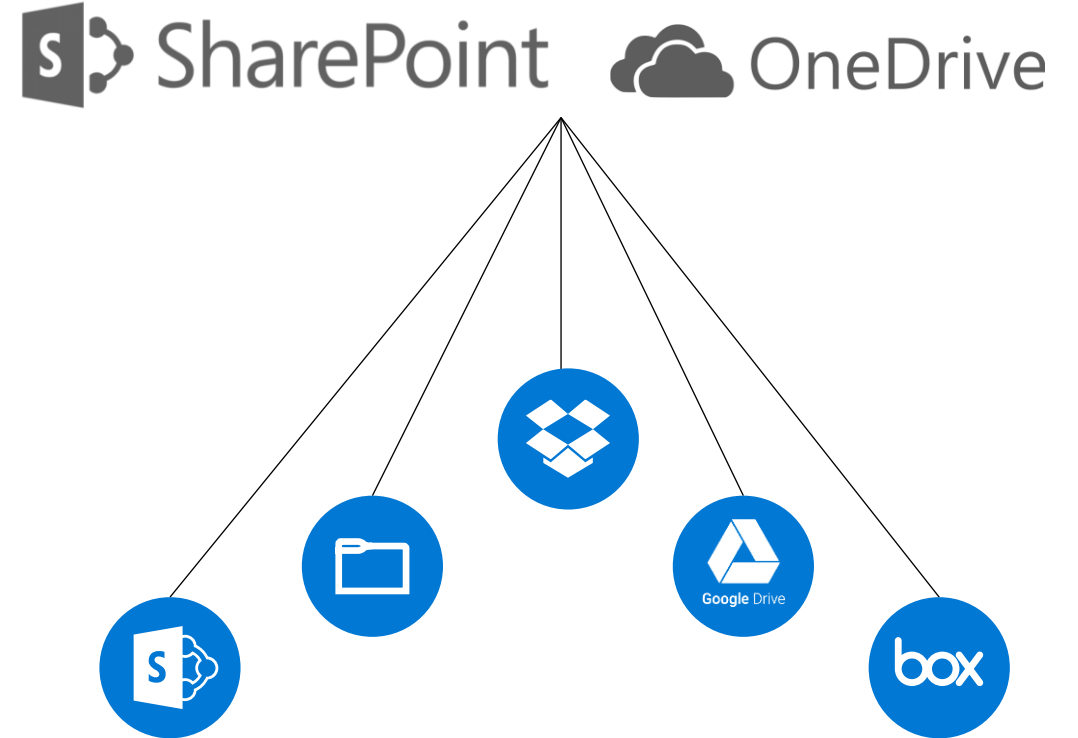
# Agenda - Chapter 1

|   |                                    |
|---|------------------------------------|
| 1 | Migration needs, options and tools |
| 2 | Introduction                       |
| 3 | Requirements and limitations       |
| 4 | How it works?                      |
| 5 | Click through                      |
| 6 | Advanced settings                  |
| 7 | Customize your migration           |

# SharePoint Migration Tool – migration needs

## Bridging the gap with migration

- Migration is the ability to move content from different sources to SharePoint and OneDrive in Microsoft 365.



# SharePoint Migration Tool - Migration options



## **Maintaining a hybrid model**

Continue to maintain hybrid model providing services on-premises or online based on the organization needs

Continue to use existing customizations on-premises

Enrich traditional on-premises scenarios with cloud innovation



## **Migration to cloud**

Migrate at your own pace to the cloud with little or no disruption to existing service

Pilot Microsoft 365 services with a subset of users

# SharePoint Migration Tool – migration tools



# Agenda - Chapter 2

|   |                                    |
|---|------------------------------------|
| 1 | Migration needs, options and tools |
| 2 | <b>Introduction</b>                |
| 3 | Requirements and limitations       |
| 4 | How it works?                      |
| 5 | Click through                      |
| 6 | Advanced settings                  |
| 7 | Customize your migration           |

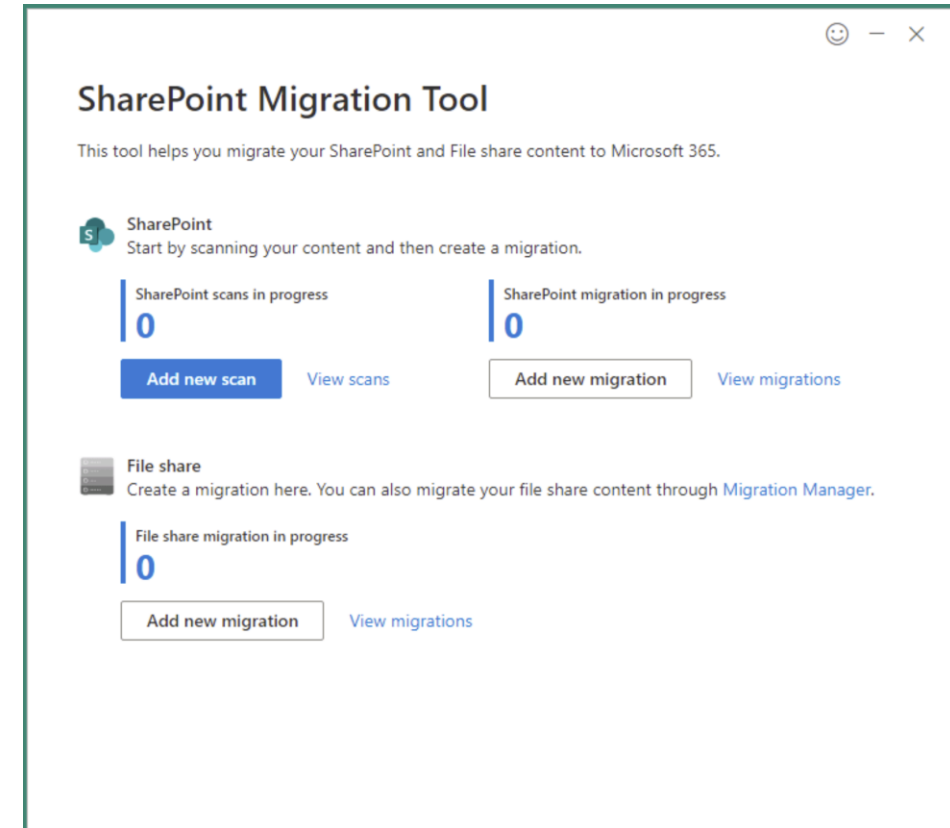
# SharePoint Migration Tool

## Free, simple, and fast

- Free, simple, and fast migration solution
- For migrating content from on-premises SharePoint sites and file shares to SharePoint or OneDrive in Microsoft 365
- It is available for all Microsoft 365 users.

## Features

- Easy to use intuitive interface
- Migrate from file shares on-premises to SharePoint Online or OneDrive for Business
- Migrate from on-premises versions of SharePoint to SharePoint Online
- Supports CSV and JSON file for bulk migrations
- Detailed information and reports to track and troubleshoot migrations
- Support for MFA accounts
- Provides PowerShell cmdlets for migration





# Install SharePoint Migration Tool

## Download and install

- You can download and install the SharePoint Migration Tool here: <https://aka.ms/spmt-ga-page>
- **There is no offline installer for this app.**



## SharePoint Migration Tool

Microsoft Corporation

Migrate content from any of these locations to SharePoint Online or OneDrive:

- sharepoint Server 2019
- SharePoint Server 2016
- SharePoint Server 2013
- SharePoint Server 2010
- Network or local file shares

☒ I agree to the [Terms of Service and Privacy Policy](#)

Install

[Install first release](#) ⓘ

New to the SharePoint Migration tool? [Learn More](#)



SharePoint Online [customer support](#)

More info: <https://docs.microsoft.com/en-us/sharepointmigration/introducing-the-sharepoint-migration-tool>

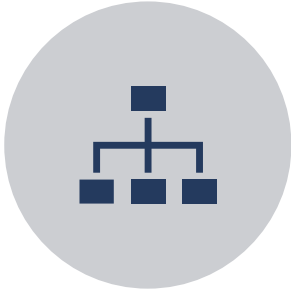
# Agenda - Chapter 3

|   |                                     |
|---|-------------------------------------|
| 1 | Migration needs, options and, tools |
| 2 | Introduction                        |
| 3 | <b>Requirements and limitations</b> |
| 4 | How it works?                       |
| 5 | Click through                       |
| 6 | Advanced settings                   |
| 7 | Customize your migration            |

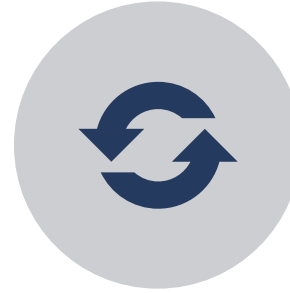
# SharePoint Migration Tool - Requirements

| Specs                   | Recommended requirements for best performance                          | Minimum requirements (expect slow performance)                         |
|-------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>CPU</b>              | 64-bit Quad core processor or better                                   | 64-bit 1.4 GHz 2-core processor or better                              |
| <b>RAM</b>              | 16 GB                                                                  | 8 GB                                                                   |
| <b>Local storage</b>    | Solid state disk: 150 GB free space                                    | Hard disk: 150 GB free space                                           |
| <b>Network card</b>     | 1 Gbps                                                                 | High speed internet connection                                         |
| <b>Operating System</b> | Windows Server 2012 R2 or<br>Windows 10 client<br>.NET Framework 4.6.2 | Windows Server 2012 R2,<br>Windows 10 or newer<br>.NET Framework 4.6.2 |

# SharePoint Migration Tool - Limitations



**No options for custom metadata mapping**



**Not supported for SharePoint on-premises to SharePoint on-premises migrations**



**Currently SharePoint 2010, 2013, 2016 and 2019 are supported**



**Not supported for SharePoint Online to SharePoint Online migrations**

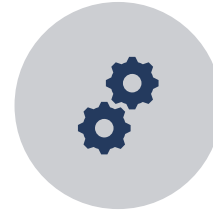
# Agenda - Chapter 4

|   |                                     |
|---|-------------------------------------|
| 1 | Migration needs, options and, tools |
| 2 | Introduction                        |
| 3 | Requirements and limitations        |
| 4 | <b>How it works?</b>                |
| 5 | Click through                       |
| 6 | Advanced settings                   |
| 7 | Customize your migration            |

# How it works?



The SharePoint Migration Tool provides a wizard-like experience..



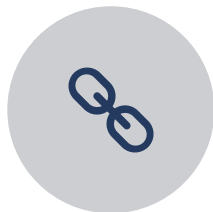
Uses Migration API behind the scene



It migrates your data *from* your on-premises SharePoint Server document libraries and local file shares *to* SharePoint Online (SPO).



Microsoft 365 Authentication happens first once you select destination (SPO/OneDrive) followed by on-premises authentication.



After selecting source and source library, you are ready to submit the Migrate Job.



The ***scanning, packaging, uploading, and importing*** steps are performed in parallel across all the files submitted for migration.

# Permissions



**You need the write permission on the destination site collection and/or OneDrive (contribute/site collection administrator)**



**You need to have the read access to source (site collection or local file share)**

# Execution phases (1/2)

1

## Authentication

- Authenticate destination tenant
- Associate submitted job with the tenant for later execution from even a different computer
- Account should be site collection admin

2

## Scanning

- A scan is performed on every file
- Scans for known potential issues
- Confirms access to source and destination

3

## Packaging

- A content package is created
- Package contains a manifest consisting of 8 XMLs



# Execution phases (2/2)

## 4

### Upload

- The data is encrypted at rest using the AES CBC 256 standard
- The files are encrypted along with the manifest files
- Package is uploaded to Azure with the manifest

## 5

### Import

- The key is provided to SPO SAS
- Only Azure and SPO are interacting to fetch and migrate the content into the destination
- This process is timer job based

# Encryption and Security



During the upload and import phases, data is encrypted, and Azure containers and keys are generated.



The default key to the container is generated programmatically and is only valid for three days.



This key is the only way to gain access to the container. It is generated randomly and not reused.



No one has direct access to the storage accounts or the containers.



Each container is dedicated to the customer who it was provided to and NOT reused.

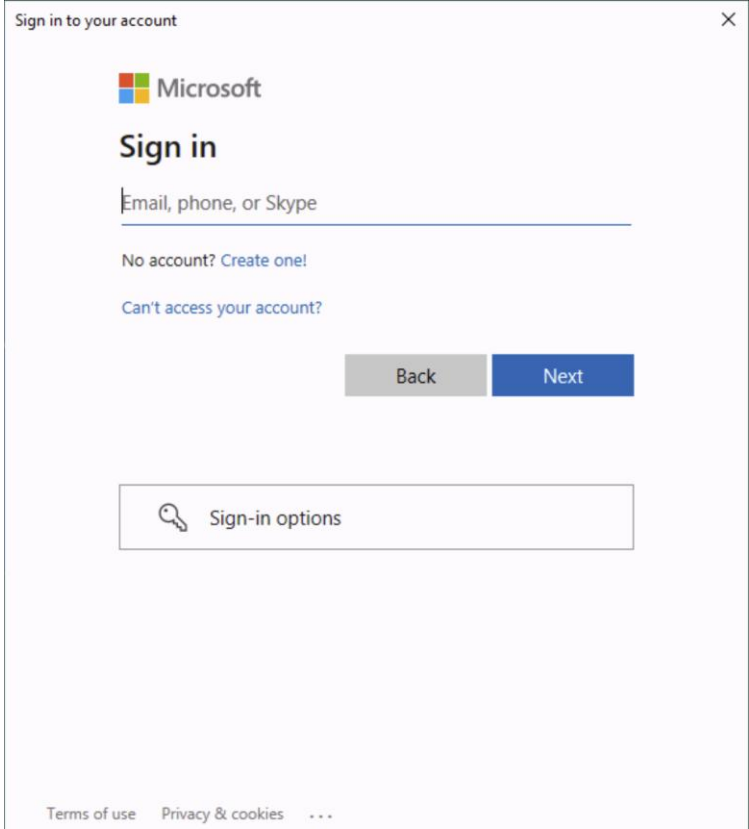


The data is stored in the Azure blob anywhere from 30 to 90 days after which it is deleted.

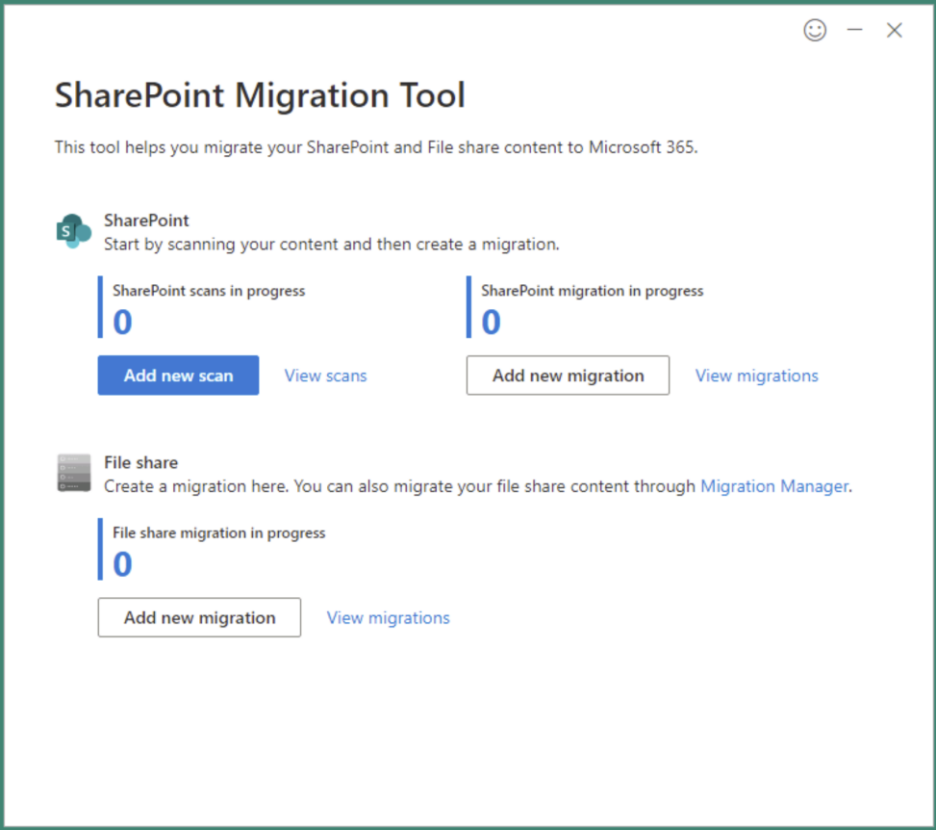
# Agenda - Chapter 5

|   |                                     |
|---|-------------------------------------|
| 1 | Migration needs, options and, tools |
| 2 | Introduction                        |
| 3 | Requirements and limitations        |
| 4 | How it works?                       |
| 5 | <b>Click through</b>                |
| 6 | Advanced settings                   |
| 7 | Customize your migration            |

# Sign in process

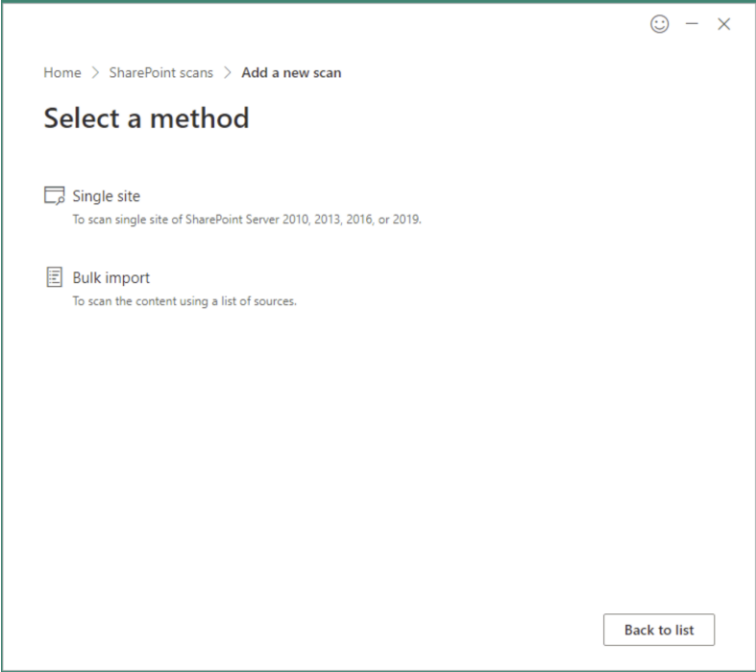


Enter Credentials

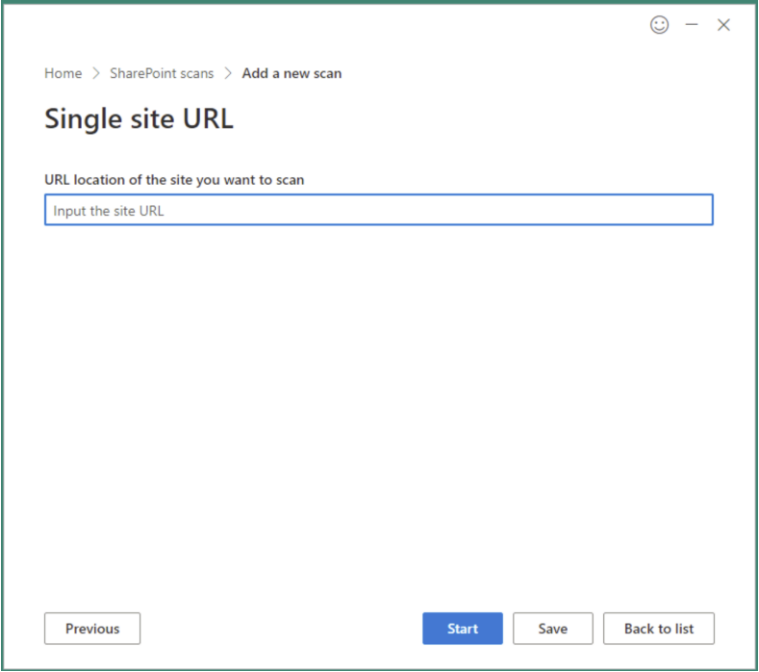


Select Source

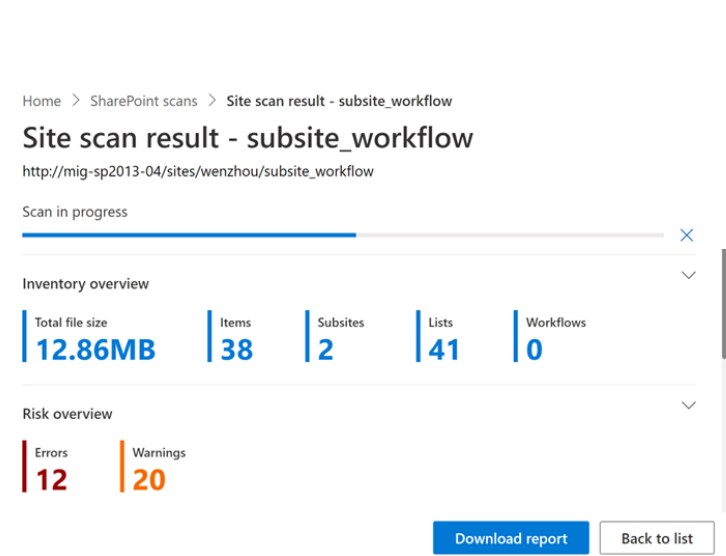
# Scan and assess a SharePoint Server site



Enter SharePoint Source URL



Specify Destination



Scan results and scan report

# SharePoint as Source – migrating a site collection (1/2)

## Select a method



### Single source URL

To migrate SharePoint Server 2010, 2013, 2016 or 2019.



### Bulk migration using JSON or CSV file

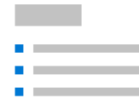
To migrate content using a list of sources and destinations.

## Select your migration type



### Site migration

Migrate your site structure and content.



### List or document library migration

Migrate your list or document library.

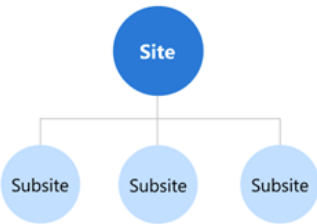


### Workflow migration

Migrate your workflows to Power Automate.

# SharePoint as Source – migrating a site collection (2/2)

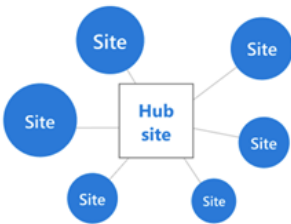
## < Select a site structure



### Keep the classic site structure

Make no changes to the current site collection structure. This choice may minimize post-migration tasks. However, you will not be able to upgrade these sites to modern sites after migration.

Select



### Switch to the modern site structure

Update all level-one subsites into modern sites, and optionally associate them with a hub. Modern sites can be connected to Microsoft 365 groups, display in global navigation, and be integrated with Microsoft Teams and Viva Connections. [Learn More](#)

RECOMMENDED

Select

## < Select hub site association

You have chosen to promote all level-one subsites to modern sites.

When you associate sites with a hub, they inherit the hub site's theme and navigation. Content from the sites will roll up to the hub site and be included in searches on the hub site. [Learn more](#)

Do you want to associate the promoted sites to a hub?

- ☒ Do not associate with a hub now
- ☐ Associate with a hub

Next Cancel

## < Modern site structure settings

You have chosen to promote all level-one subsites to modern sites.

When you associate sites with a hub, they inherit the hub site's theme and navigation. Content from the sites will roll up to the hub site and be included in searches on the hub site. [Learn more](#)

Do you want to associate the promoted sites to a hub?

- ☐ Do not associate with a hub now
- ☒ Associate with a hub

☒ Register your destination as a hub site

Destination site  
<https://291020201130.sharepoint.com/example>

Hub name

☐ Associate with an existing hub site

# File Share as Source

Select a source and destination

What file share do you want to migrate?  
C:\Users\uafzal\Desktop\IPWork\ref

Select a destination SharePoint site  
Enter a URL  
https://pfeusman.sharepoint.com/sites/usman-tub-demos

Choose a document library  
From Usman-Tub-demos

Documents

Documents

Focused Content

Form Templates

Migrated Content

PersonalDocumentLib

Preservation Hold Library

Project Documents

Restricted Document Library

Site Assets

Site Pages

Style Library

Select Destination Site

Select a source and destination

What file share do you want to migrate?  
C:\Users\uafzal\Desktop\IPWork\ref

Select a destination SharePoint site  
Enter a URL  
https://pfeusman-my.sharepoint.com/personal/admin\_usmanafzal\_com/

Choose a document library  
From mod administrator

Documents

Documents

Form Templates

Style Library

Back

Add

Cancel

Select Destination Library

SharePoint Migration Tool - v0.2.70.0

Choose a source and pick your destination

| Source       | Destination       |   |
|--------------|-------------------|---|
| ▼ C:\...\ref | Usman-Tub-demos   | × |
| ref          | Migrated Content  |   |
| ▼ C:\...\ref | mod administrator | × |
| ref          | Documents         |   |

+ Add a task

Save tasks to CSV

Back

Migrate

Cancel

Review Job Config



# CSV as Source

Select a source and destination

Select a CSV for bulk migration

C:\Users\ufzal\Desktop\MigrateTasks.csv

Choose File

Back

Add

Cancel

Specify CSV

SharePoint Migration Tool - v0.2.70.0

Choose a source and pick your destination

| Source       | Destination       |   |
|--------------|-------------------|---|
| ▼ C:\...\ref | Usman-Tub-demos   | × |
| ref          | Migrated Content  |   |
| ▼ C:\...\ref | mod administrator | × |
| ref          | Documents         |   |

+ Add a task

Save tasks to CSV

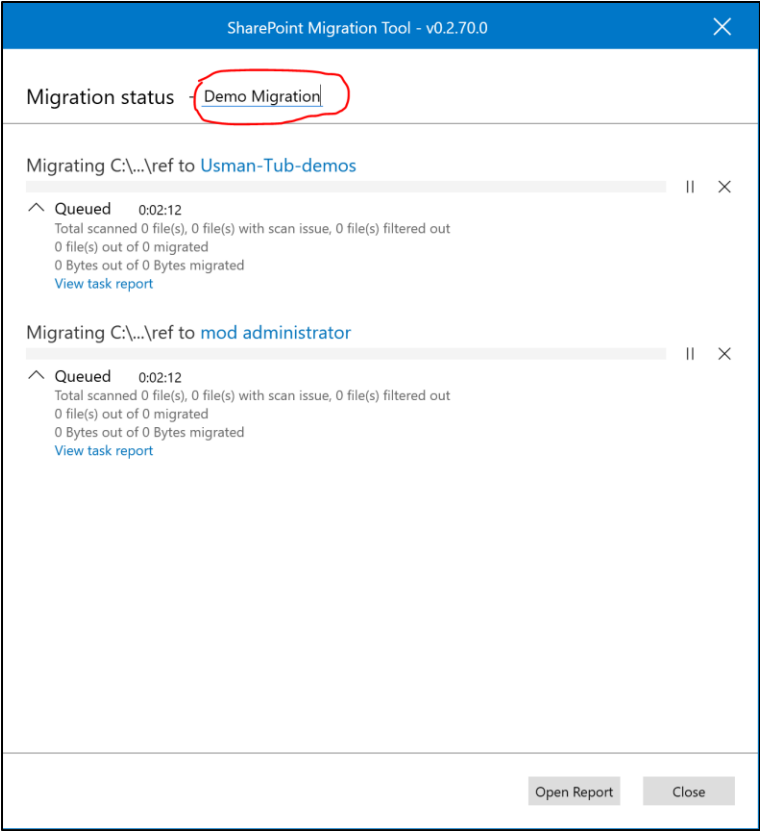
Back

Migrate

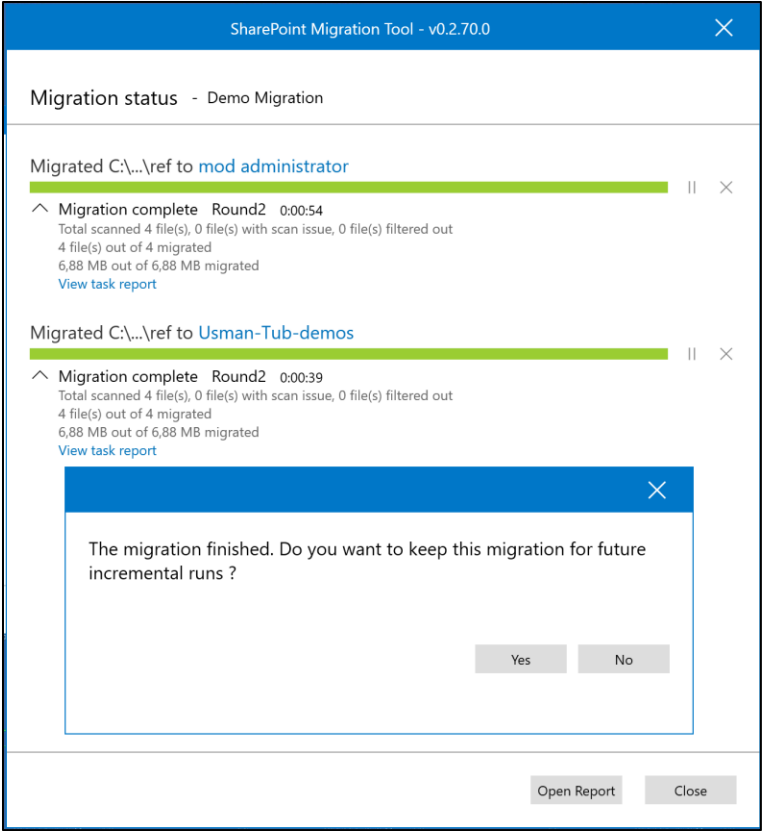
Cancel

Review Job Config

# Migration Progress



Review Progress



Job Completed

# Migrating Permissions

|                                                                                                             | File share                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SharePoint on-premises files                                                                              |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| User is mapped between on-premises and SPO (either Dirsync has been run or a user mapping file is provided) | <ul style="list-style-type: none"><li>• There are only two types of permissions that will be migrated: Read and Write.</li><li>• If a file has <b>Write</b> permission for User1, then the file will be set to Contribute for User1 in SPO.</li><li>• If a file has <b>Read</b> permission for User1, then the file will be set to Read for User1 in SPO.</li><li>• NOTE: At this time, the special permissions, such as Deny, will not be saved.</li></ul> | All the unique permissions on a file will be migrated to SPO. Inherited permissions will not be migrated. |
| No user mapping (not synced, no user mapping file)                                                          | Files will be assigned the default permission of the location to which it has been migrated in SPO.                                                                                                                                                                                                                                                                                                                                                         | Files will be assigned the default permission of the location to which it has been migrated in SPO.       |

# Agenda - Chapter 6

|          |                                     |
|----------|-------------------------------------|
| 1        | Migration needs, options and, tools |
| 2        | Introduction                        |
| 3        | Requirements and limitations        |
| 4        | How it works?                       |
| 5        | Click through                       |
| <b>6</b> | <b>Advanced settings</b>            |
| 7        | Customize your migration            |

# Advanced settings



## Only perform scanning

If you wish to scan the files as a pre-assessment to migration, turn on **Only perform scanning**.



## Enable incremental migration

It lets you rerun the migration jobs at a later date, migrating only the changes or additions since the previous run.



## Migrate file version history

If set to **No**, only the most recent version of the file will be migrated.  
  
If set to **Yes**, you can choose whether to keep all versions or limit it to a specific number.



## Do not migrate hidden files

If turned **on**, hidden system files will not be migrated.

Settings

Only perform scanning

☐ Off

Enable incremental migration

☒ On

Migrate file version history

☒ On

Keep all versions

☒ On

Do not migrate hidden files

☐ Off

Do not migrate files created before

Only applies to SharePoint on-premises files

Select a date

15

Do not migrate files modified before

Applies to both SharePoint on-premises files and local file shares

Select a date

15

Do not migrate files with these extensions

Use ';' to separate multiple extensions. e.g. mp4;avi;mkv

Do not migrate files and folders with invalid characters

☒ On

Save

Cancel

# Advanced settings (Contd.)



## Do not migrate files created before

If you choose to limit which files are migration based on creation date, set your values in this section.

This may be to limit the number of files migrated or to adhere to overall company governance policy regarding file retention.



## Do not migrate files modified before

If you choose to limit what files are migration based on modified date, set your values in this section.

This may be to limit the number of files migrated or to adhere to overall company governance policy regarding file retention.



## Do not migrate files with these extensions

To prevent certain file types from migrating, list each extension, separating it with a vertical bar. For example, mp4|avi|mkv. Do not include the leading "." before the extension name.

Settings

Do not migrate files and folders with invalid characters

☐ On

Migrate OneNote folder as OneNote notebook

☐ Off

Azure Active Directory lookup

☒ On

Preserve user permissions

☒ On

User mapping file

Choose File

Default mapping will be used if no file is provided | [Learn more](#)

SharePoint Migration Tool working folder

Choose Folder

Temp folder will be used by default

Use custom Azure storage

☐ Off

Version 0.3.70.0

Save

Cancel

# Advanced settings (Contd. 2)



## Do not migrate files and folders with invalid characters

If set to **On**, the tool will skip any potential special characters. While this can improve performance when the source potentially contains a high number of files containing invalid characters, it also has drawbacks.



## Azure Active Directory lookup

By default, this is set to **On**. If no User mapping file is provided by the user, then Azure Active Directory is used as the default for user mapping.



## Preserve user permissions

By default, this is set to **On**. If set to Off, no permissions will be preserved.

Settings

Do not migrate files and folders with invalid characters

☐ On

Migrate OneNote folder as OneNote notebook

☐ Off

Azure Active Directory lookup

☒ On

Preserve user permissions

☒ On

User mapping file

Choose File

Default mapping will be used if no file is provided | [Learn more](#)

SharePoint Migration Tool working folder

Choose Folder

Temp folder will be used by default

Use custom Azure storage

☐ Off

Version 0.2.70.0

Save

Cancel

# Advanced settings (Contd. 4)



## User mapping file

If you wish to use your own mapping file, select the file to be used by clicking **Choose file**.



## SharePoint Migration Tool working folder

By default, a temp folder will be created. If you wish to specify a specific working folder, enter the name here.



## Use custom Azure storage

If you wish to use your own Azure storage, set this value to On.

Settings

Do not migrate files and folders with invalid characters

☐ On

Migrate OneNote folder as OneNote notebook

☐ Off

Azure Active Directory lookup

☒ On

Preserve user permissions

☒ On

User mapping file

Choose File

Default mapping will be used if no file is provided | [Learn more](#)

SharePoint Migration Tool working folder

Choose Folder

Temp folder will be used by default

Use custom Azure storage

☐ Off

Version 0.2.70.0

Save

Cancel



# Agenda - Chapter 7

|   |                                     |
|---|-------------------------------------|
| 1 | Migration needs, options and, tools |
| 2 | Introduction                        |
| 3 | Requirements and limitations        |
| 4 | How it works?                       |
| 5 | Click through                       |
| 6 | Advanced settings                   |
| 7 | <b>Customize your migration</b>     |

# Customize migration

## Using a comma separated value (CSV) file for data content migration

- You can use a comma separated (CSV) file to bulk migrate your data.
- There are six columns needed in your CSV file
  - The first three are your **source values**, each providing detail about where your data is currently located.
  - The remaining three columns indicate the **destination** - the site, document library and optional subfolder to where you are migrating your data.
  - All six columns must be accounted for in the file, even if you are not needing a value for a given field.

|   | A                                                 | B                 | C                 | D                                            | E              | F                 |
|---|---------------------------------------------------|-------------------|-------------------|----------------------------------------------|----------------|-------------------|
| 1 | C:\MigrationTests\testFiles                       |                   |                   | https://contoso.sharepoint.com/sites/Sample/ | DocLibraryName | OptionalSubfolder |
| 2 | https://sharepoint2013.com/sites/contosoteamsite/ | Source DocLibrary | OptionalSubfolder | https://contoso.sharepoint.com/sites/test    | Documents      | OptionalSubfolder |

# Customize migration: Creating CSV

## Using a comma separated value (CSV) file for data content migration

- This example shows how it would appear in a text file:

### Format:

Source,SourceDocLib,SourceSubFolder,TargetWeb,TargetDocLib,TargetSubFolder

### Example: File Share to SPO

C:\MigrationTests\testfiles,,,https://contoso.sharepoint.com/sites/Sample/,DocLibrary  
Name,DocLibraryName\_subfolder

### Example: SharePoint on-premises to SPO

https://sharepoint2013.com/sites/contosoteamsite/,DocumentLibraryName,DocLibrarySubfo  
lder\_name,https://contoso.sharepoint.com/sites/Sample/,DocLibraryName,DocLibraryName\_  
subfolder

**Important:** If you use the standard out-of-the-box Document library ("Shared Documents"), you must use the internal name "Documents" as the placeholder value for the Source Document Library (Column B) OR your Target Document Library (Column E) in your CSV file. If you enter "Shared Documents" in that column, you will receive an "invalid document library" error.

# Customize migration: CSV fields

| Field            | Description                                                                                                                                                                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source           | Required: Enter either a file share path or an on-premises SharePoint Server site URL.                                                                                                                                                                                                                                      |
| Source DocLib    | Optional: Enter name of the SharePoint Server document library you are migrating. If you leave this field empty, all document libraries will be migrated. If you are migrating a file share, leave this column empty.                                                                                                       |
| Source SubFolder | Optional: Enter the name of the subfolder in the document library. If this column is left empty, the migration starts from the root. If there is a value in this column, the migration starts from the subfolder and down.<br>This column is used only for SharePoint Server sites. It is ignored for file share migration. |
| Target Web       | Required: Enter the SharePoint Online site URL where the files are to be migrated.                                                                                                                                                                                                                                          |
| Target DocLib    | Required: Enter the name of the document library with the SharePoint Online site where the files are to be migrated.                                                                                                                                                                                                        |
| Target SubFolder | Optional: Enter the name of the subfolder in the document library. If this column is left, then empty the files which will be moved to the root level.                                                                                                                                                                      |

# Knowledge Check

**Name the three sources for migration using SharePoint Migration Tool?**

- SharePoint on-premises
- File shares
- CSV file with source information

**What version of SharePoint on-premises are supported?**

- SharePoint 2010, 2013, 2016 and 2019

Questions?



